NBA Shot Make Probability Prediction Project

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Introduction

The objective of this project was to predict the probability that an NBA shot would be made using player tracking and shot-context data. The training dataset contained historical shot attempts and outcomes, while the testing dataset contained the same shot characteristics without outcomes. The goal was to develop a model capable of accurately estimating shot make probability for unseen observations.

To accomplish this, I first explored the data and developed a baseline logistic regression model. I then transitioned to an XGBoost model, which was better suited for capturing nonlinear relationships and interactions among variables. A major focus of the project was feature engineering using basketball-specific concepts to improve predictive performance and better represent shot quality.

Data and Methodology

The dataset contained information on shot location, shot type, defender proximity, number of contesting defenders, shot clock, game state, player movement, and other contextual variables.

The initial modeling process focused on variables that intuitively influence shot success, including:

- Shot distance
- Shot type
- Defender distance
- Number of contesting defenders
- Shot location
- Shot clock
- Shooter speed
- Game state

After developing a baseline model, I implemented XGBoost because of its ability to model complex interactions among features and improve predictive accuracy.

Feature Engineering

One of the primary goals of this project was to create basketball-specific features beyond those directly provided in the dataset.

Contester Density

The most impactful engineered feature was a custom defensive pressure metric:

$$ContesterDensity = \frac{NumContesters}{ClosestDefenderDistance}$$

This metric combines two important aspects of defense:

- The number of defenders contesting the shot
- How close the nearest defender is to the shooter

For example, a shot with three defenders contesting from close range should be considered more difficult than a shot with one nearby defender, even if the nearest defender distance is identical.

The resulting feature became the **third most important predictor** in the final model.

Player Field Goal Percentage

I also created a shooter-specific effectiveness metric.

Each player's historical field goal percentage was calculated using the training dataset and assigned as a new feature (player_fg).

This allowed the model to account for differences in shooting ability among players rather than assuming all shooters perform equally.

This feature became the **sixth most important variable** in the final model.

Closeout Speed

The dataset included defender approach trajectories leading into each shot.

Using this information, I created a custom metric measuring how aggressively a defender closed out on a shooter (closeout_speed).

This feature captured information not fully represented by static defender distance alone and ranked among the most important engineered variables in the final model.

Additional Features

Additional engineered variables included:

- Rim Shot Indicator
- Catch-and-Shoot Indicator
- Late Shot Clock Indicator

Several additional features were tested and ultimately removed after feature importance analysis showed limited predictive value.

Model Development

The final model was built using XGBoost.

XGBoost was selected because it can effectively learn nonlinear relationships and interactions among variables that traditional linear models may not capture.

Variables included in the final model were:

- Distance
- Shot Type
- Contester Density
- Closest Defender Distance
- Shot Location (X and Y Coordinates)
- Player FG%
- Shooter Speed
- Shot Clock
- Closeout Speed

- Catch-and-Shoot Indicator
- Late Clock Indicator
- Game State

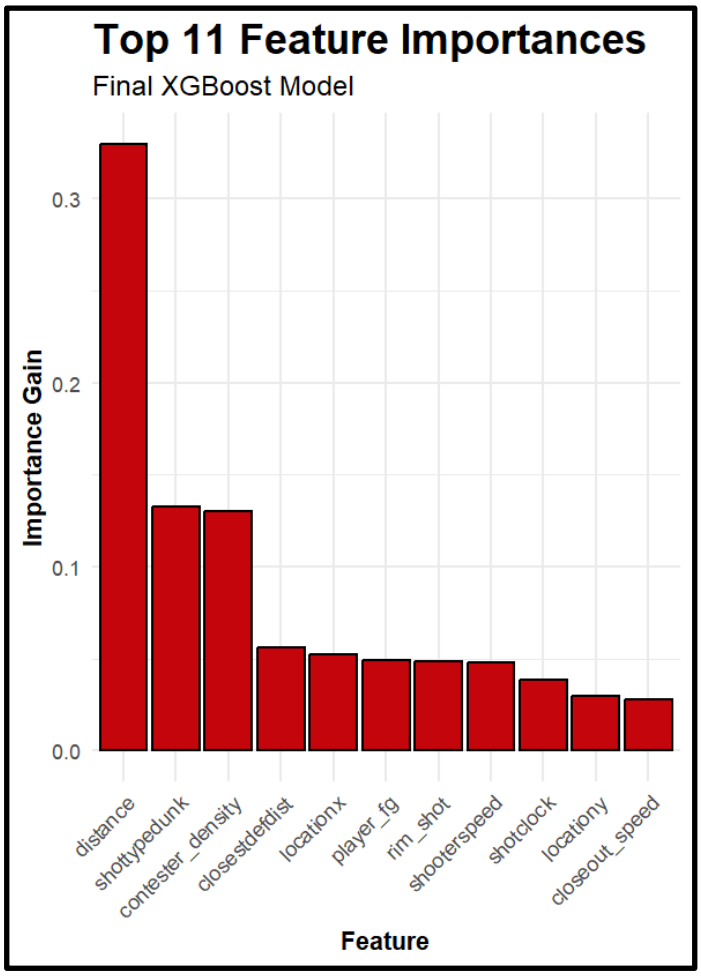
Features that contributed minimal information were removed to simplify the model and improve interpretability.

Results

Feature Importance

The most important variables in the final model were:

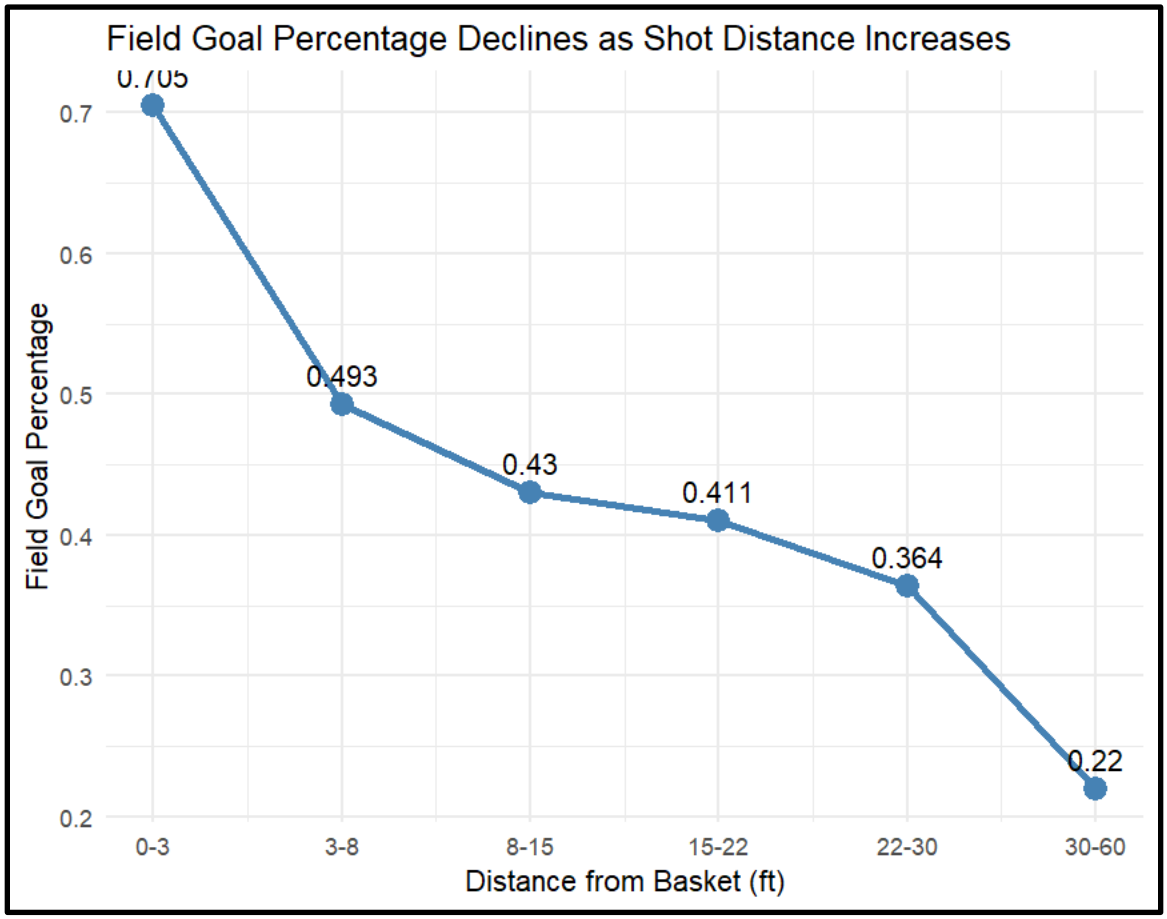
Feature	Rank
Distance	1
Shot Type (Dunk)	2
Contester Density	3
Closest Defender Distance	4
Location X	5
Player FG%	6
Rim Shot	7
Shooter Speed	8
Shot Clock	9
Location Y	10
Closeout Speed	11



The feature importance results indicate that shot quality is primarily driven by distance, shot type, defensive pressure, and shooter ability.

Impact of Shot Distance

Distance was the most important variable in the model.

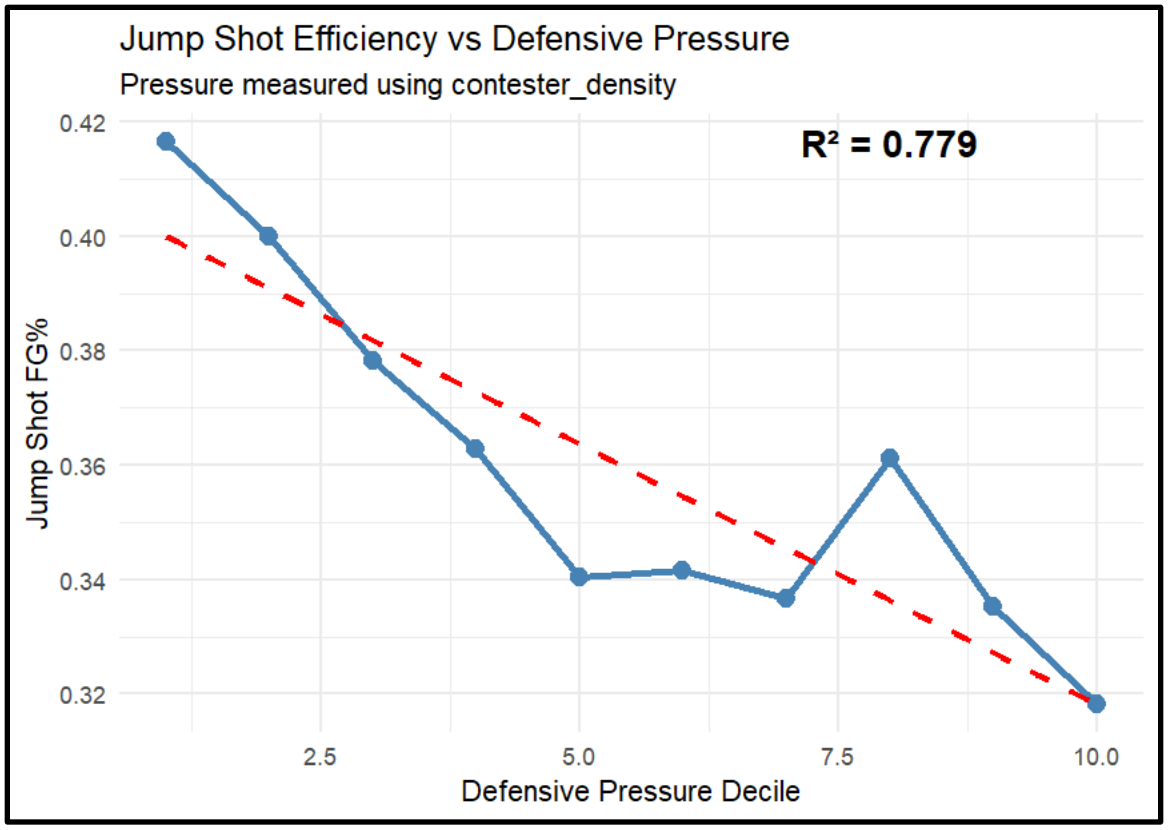


The chart demonstrates a clear relationship between distance and shooting efficiency. Field goal percentage declined steadily as shot distance increased, from approximately 70.5% on shots within three feet of the basket to approximately 22.0% on shots beyond thirty feet.

This finding aligns with basketball intuition and explains why distance emerged as the most influential variable in the model.

Impact of Defensive Pressure

To isolate the effect of defensive pressure, I focused specifically on jump shots and grouped observations into defensive-pressure deciles using the custom contester_density metric.



Jump-shot efficiency declined from approximately 42% in the lowest-pressure decile to approximately 32% in the highest-pressure decile. The relationship exhibited a strong negative trend ($R^2 = 0.779$).

This finding helps explain why contester_density ranked as the third most important feature in the final model and demonstrates the value of combining defender proximity and defender volume into a single defensive-pressure metric.

Key Findings

Several conclusions emerged from the analysis:

1. Shot distance was the strongest predictor of shot success.
2. Defensive pressure significantly reduced shooting efficiency.
3. The custom contester_density metric was more informative than considering defender distance or defender count individually.
4. Shooter skill remained important even after controlling for shot quality and defensive context.

5. Defender closeout behavior provided additional predictive information beyond static defender distance.

Reflection

This project represented my first significant experience using coding and machine learning tools to solve an analytics problem. It was surprising to see how fast I picked up programming knowledge through the leverage of artificial intelligence. I am incredibly excited to start new projects and see what I can create.

The most valuable aspect of the project for me personally was the discovery. I never realized the capabilities of these programming languages. I never realized how fast AI could teach me and help materialize my ideas. There are ideas of sports analytics models that are floating in my mind. Now I have the ability and knowledge to create them.

Future Improvements

Given additional time, I would explore:

- Shooter effectiveness by shot type
- Team-level offensive and defensive metrics
- Additional defensive matchup features
- Cross-validation and hyperparameter optimization
- Spatial shot-quality metrics incorporating court zones

Conclusion

The final XGBoost model successfully generated shot-make probabilities for all observations in the testing dataset. Through feature engineering and basketball-specific analysis, several meaningful predictive variables were identified. In particular, the custom contester_density, player_fg, and closeout_speed features provided substantial predictive value and improved the model's ability to capture defensive pressure, player skill, and contest quality.

The project reinforced the importance of combining domain knowledge with data-driven modeling techniques to generate stronger and more interpretable predictive models.